

SOC Detection Catalog

Splunk detection logic, thresholds, ATT&CK; mapping, false positives, tuning, and escalation criteria

Portfolio Scope

The detections below use synthetic portfolio data and are designed to demonstrate junior SOC detection and triage methodology. Thresholds are starting points for a lab; production thresholds should be baselined and tuned to the environment.

Detection	Data source / SPL file	Trigger	Severity	ATT&CK;	False positives / tuning	Escalate when
Brute Force	Authentication logs / 01_brute_force.spl	>=10 failures from one source in 5 min	Medium; High at >=20	T1110 - Brute Force	User typo loops, testing, scanners. Exclude approved scanners; baseline service accounts.	Failures are followed by success, privileged user targeted, or multiple accounts are impacted.
Password Spray	Authentication logs / 02_password_spray.spl	>=5 users from one source in 10 min	High	T1110.003 - Password Spraying	Security testing, shared proxy, SSO health checks. Allowlist validated test sources only.	Any success occurs, privileged account is targeted, or source persists across windows.
Credential Stuffing	Authentication logs / 03_credential_stuffing.spl	>=8 failures across >=3 users plus >=1 success in 10 min	High	T1110.004 - Credential Stuffing	QA automation or shared NAT. Correlate user agents/device/MFA before closing.	Successful login has unfamiliar device, weak auth, abnormal geo, or follow-on activity.
Failed -> Success	Authentication logs / 04_failed_then_success.spl	Failure sequence followed by success from same user/source within 10 min	High	Credential Access / Valid Account follow-on	Legitimate user recovering from mistyped password. Check MFA/device history.	Device/geo is new, MFA absent, privileged account involved, or post-login anomalies appear.
Impossible Travel	Authentication logs / 05_impossible_travel.spl	Successful logins from different geographies <=60 min apart	High	Valid Accounts concept	VPN, mobile carrier routing, corporate egress. Tune with known VPN ranges and device trust.	Unmanaged device, password-only login, risk IP, or suspicious post-authentication activity.
Suspicious PowerShell	Endpoint process telemetry / 06_suspicious_powershell.spl	Encoded or download-oriented PowerShell patterns	High	T1059.001 - PowerShell	Admin automation, software deployment. Tune signed scripts, approved parents, known paths.	Office/browser parent, network connection, persistence, credential access, or unsigned script behavior is present.

Detection Engineering Quality Standard

- Define the security behavior and required data before writing the search.
- Use a measurable trigger: count, distinct users, time window, success/failure sequence, or behavior pattern.
- Assign severity from business impact and evidence, not the alert name alone.
- Document benign scenarios and tuning options before promoting a rule.
- Test with known-positive and known-benign examples.
- Record required triage fields so another analyst can reproduce the decision.
- Track false positives and missed detections as feedback for rule improvement.

Disposition Vocabulary

- **True Positive - Escalated:** Malicious or unauthorized activity confirmed and requires higher-tier response.
- **True Positive - Contained:** Malicious activity confirmed and immediate containment is complete.
- **Benign Positive:** Rule correctly matched behavior that is authorized or expected.
- **False Positive:** Detection logic incorrectly classified non-target behavior; tuning is required.
- **Informational:** Useful context with no response action required.